# ENVS 193DS Homework 3

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## Part 1: Set up

```
# reading in packages
library(tidyverse)
library(janitor)
library(here)
library(readxl)

# reading in data
# kelp data
kelp <- read.csv(
  here("data", "temp-kelp.csv"))
# personal data
bike_rides <- read_xlsx(
  here("data", "bike-rides.xlsx"))
```

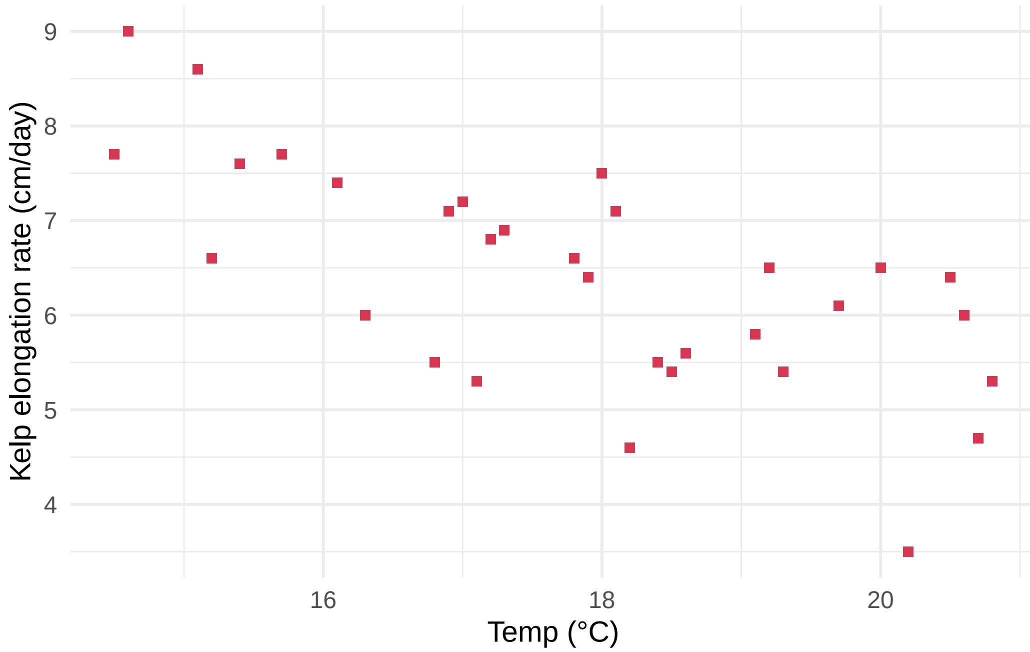
## Part 2: Problems

### Problem 1

a. Pearson's correlation and Spearman rank correlation are two tests that determine the strength of the relationship between temperature and giant kelp frond elongation rate. Pearson's correlation requires continuous, normally distributed variables with a linear relationship, while Spearman rank correlation does not require normally distributed variables.

b.

```r
# base layer ggplot
ggplot(data = kelp,
       mapping = aes(x = temp_c, # x axis is temp
                     y = kelp_elong)) + # y axis is kelp growth rate
  # first layer: points
  geom_point(color = "#D6364F", # custom color
             shape = 15) + # make points square
  # rename axes, add units
  labs(x = "Temp (\u00B0C)",
       y = "Kelp elongation rate (cm/day)") +
  theme_minimal() # custom theme
```

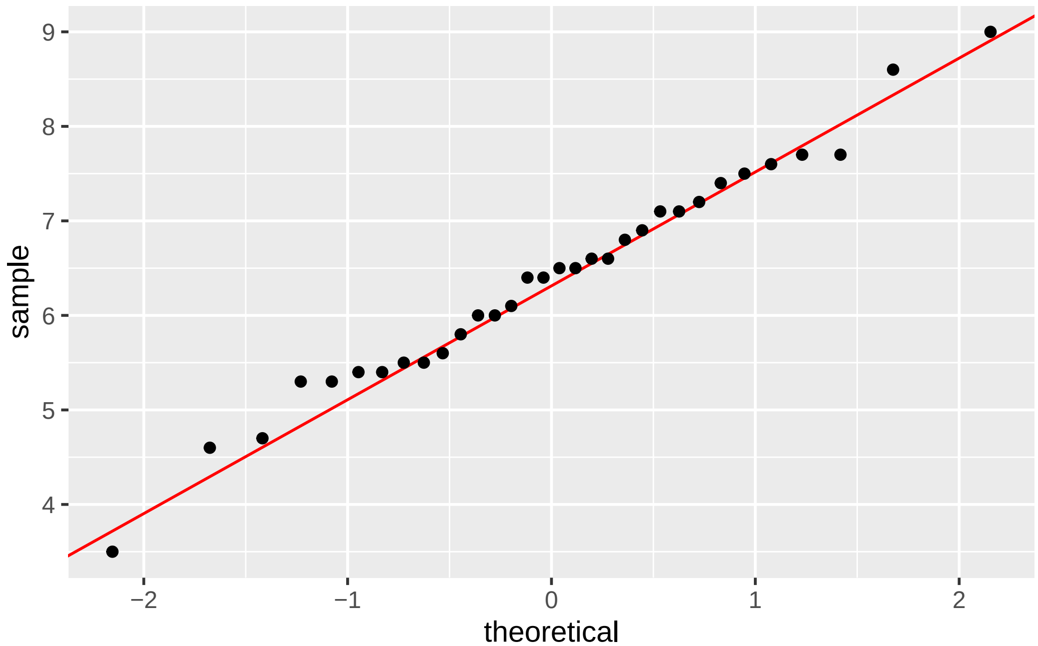



c.

**Check assumptions**

```r
# base layer: ggplot
ggplot(data = kelp, # starting data frame
aes(sample = kelp_elong)) + # y-axis for QQ plot
# first layer: QQ reference line
geom_qq_line(color = "red") + # adding a color
```

```
# second layer: QQ plot
geom_qq()
```




I checked the assumption of normality by creating a QQ plot. No curved pattern in the data is evident on the plt, meaning the data is approximately normally distributed.

**Run the test**

```
# run pearson's correlation test
cor.test(kelp$kelp_elong, kelp$temp_c, # specify variables within data frame
         method = "pearson")
```

```

	Pearson's product-moment correlation

data:  kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
```

```
sample estimates:
       cor
-0.6877489
```

d. I used a Pearson's correlation test to evaluate the strength of the relationship between temperature and giant kelp frond elongation rate because the data is continuous, normally distributed, and has a linear relationship.

We found a moderate negative relationship between temperature and giant kelp frond elongation rate (Pearson's r = -0.69, t(30) = -5.2, $p < 0.001$, $\alpha = 0.05$). As temperature increases, giant kelp frond elongation rate decreases.

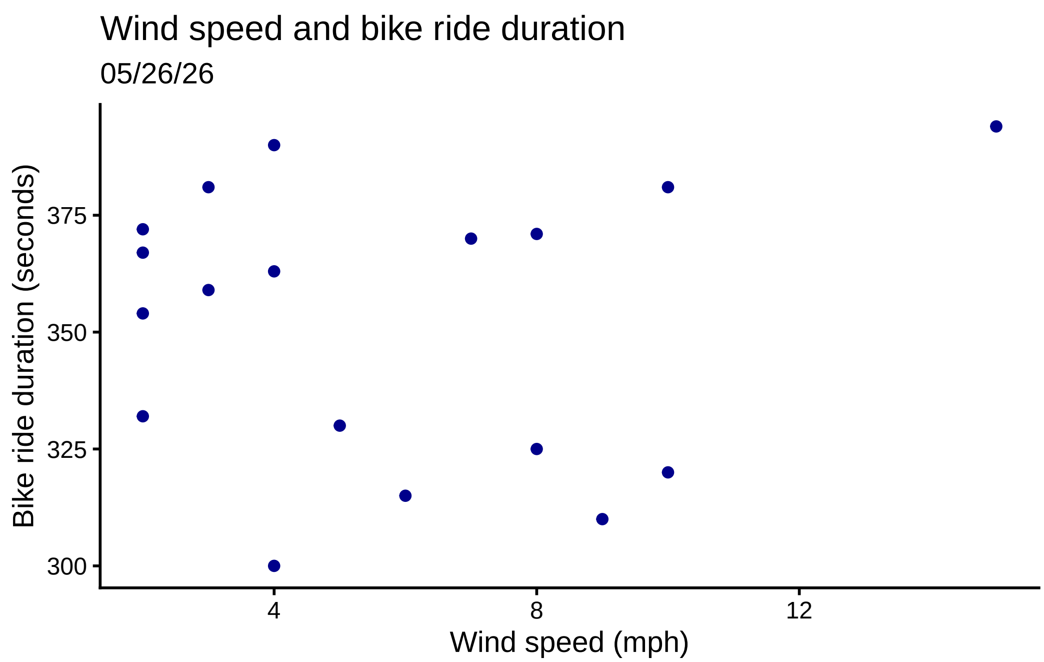
e. Lower temperatures are conducive for increased giant kelp growth rates

## Problem 2

a.

```
# cleaning data
wind_speed_and_duration <- bike_rides |> # create new object from rides
   clean_names() |> # clean column names
  # select wind speed and duration columns
  select(wind_speed_mph, duration_seconds)

# plot the data, base layer ggplot
ggplot(data = wind_speed_and_duration,
       # assign x as wind speed, y as ride duration
       mapping = aes(x = wind_speed_mph,
                     y = duration_seconds,
                     color = "wind_speed_and_duration")) + # add color
  # first layer: points
  geom_point() +
  # manually assign colors
  scale_color_manual(values = c("wind_speed_and_duration" = "darkblue")) +
    # rename x and y axes, add descriptive title and subtitle
  labs(title = "Wind speed and bike ride duration",
       subtitle = "05/26/26",
       x = "Wind speed (mph)",
       y = "Bike ride duration (seconds)") +
  theme_classic() + # custom theme
  # taking out legend
  theme(legend.position = "none")
```



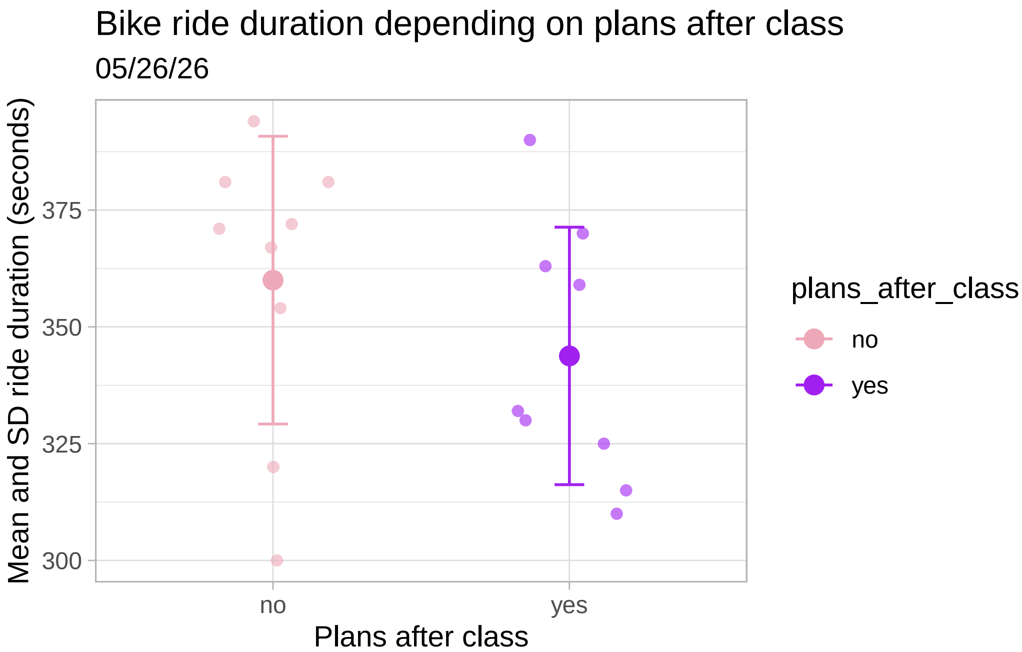

```r
# cleaning data
plans_and_duration <- bike_rides |> # create new object from rides
   clean_names() |> # clean column names
  # select plans and duration columns
  select(plans_after_class, duration_seconds)

# create new object from plans_and_duration
plans_and_duration_summary <- plans_and_duration |>
  group_by(plans_after_class) |>
  summarize( # what I want in the summary
    mean = mean(duration_seconds),
    n = length(duration_seconds),
    sd = sd(duration_seconds))

plans_and_duration_summary # display summary
```

```
# A tibble: 2 x 4
  plans_after_class  mean     n    sd
  <chr>             <dbl> <int> <dbl>
1 no                 360      9  30.8
2 yes                344.     9  27.6
```

```r
# base layer ggplot
ggplot(data = plans_and_duration,
       # x-axis should be plans after class
       mapping = aes(x = plans_after_class,
                     y = duration_seconds, # y-axis is ride duration
                     color = plans_after_class)) + # color by plans status
  # first layer: showing the underlying data
  geom_jitter(height = 0, # no jitter in the vertical direction
              width = 0.2, # smaller jitter in the horizontal direction
              alpha = 0.6) + # make the points more transparent
  # second layer: represent means at each site
  stat_summary(geom = "point",
               fun = mean,
               size = 3) + # make point bigger
  # third layer: bars to represent standard deviation
  geom_errorbar(data = plans_and_duration_summary, # use summary data frame
                mapping = aes(x = plans_after_class,
                              y = mean, # assign axes based on data frame
                              ymin = mean - sd,
                              ymax = mean + sd),
                width = 0.1) + # make the bars narrower
  # relabelling x- and y-axes, adding descriptive title and subtitle
  labs(title = "Bike ride duration depending on plans after class",
       subtitle = "05/26/26",
       x = "Plans after class",
       y = "Mean and SD ride duration (seconds)") +
  theme_light() + # custom theme
  scale_color_manual(values = c("yes" = "purple", # custom colors
                                "no" = "pink2"))
```




b. Figure 1 (continuous): Points represent individual bike rides.

Figure 2 (categorical): Data manually recorded. Small transparent points represent individual bike rides (purple: plans after class, pink: no plans after class).

### Problem 3

a. For my data, an affective visualization could look like

b.

### Problem 4

a.